

Partial Differential Equations 3 (Numerical)

Lecture 8 - Academic Year 2025/2026

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Aims and Objectives

1. To finish discussion on Thomas Algorithm.
2. To discuss Crank-Nicolson scheme in 2D.
3. To explain Alternating Direction Implicit (ADI) method.
4. Introduction to Hyperbolic PDEs.

Thomas Algorithm

Note: We will finish our discussion on Thomas algorithm in Lecture 8. For convenience, I repeat the section on Thomas Algorithm from Lecture 7 here.

A good algorithm to solve the matrix equation for 1D heat equation is the Thomas algorithm. This is effective when we have a tri-diagonal matrix, that is a matrix where all the elements are zero except for those in the main diagonal, the subdiagonal/lower diagonal (the first diagonal below), and the supradiagonal/upper diagonal (the first diagonal above). The key idea behind the Thomas algorithm is to turn the matrix equation in Eq. (1)

$$\begin{bmatrix} b_1 & c_1 & & & & \\ a_2 & b_2 & c_2 & & & \\ & a_3 & b_3 & c_3 & & \\ & & & \dots & & \\ & & & & a_{n_x-4} & b_{n_x-4} & c_{n_x-4} & 0 \\ & & & & & a_{n_x-3} & b_{n_x-3} & c_{n_x-3} \\ & & & & & & a_{n_x-2} & b_{n_x-2} \end{bmatrix} \begin{bmatrix} T_1^{n+1} \\ T_2^{n+1} \\ T_3^{n+1} \\ \dots \\ T_{n_x-4}^{n+1} \\ T_{n_x-3}^{n+1} \\ T_{n_x-2}^{n+1} \end{bmatrix} = \begin{bmatrix} d_1^n \\ d_2^n \\ d_3^n \\ \dots \\ d_{n_x-4}^n \\ d_{n_x-3}^n \\ d_{n_x-2}^n \end{bmatrix} \quad (1)$$

to a form that looks like

$$\begin{bmatrix} 1 & \beta_1 & & & & \\ 0 & 1 & \beta_2 & & & \\ & 0 & 1 & \beta_3 & & \\ & & & \dots & & \\ & & & & 0 & 1 & \beta_{n_x-4} \\ & & & & & 0 & 1 & \beta_{n_x-3} \\ & & & & & & 0 & 1 \end{bmatrix} \begin{bmatrix} T_1^{n+1} \\ T_2^{n+1} \\ T_3^{n+1} \\ \dots \\ T_{n_x-4}^{n+1} \\ T_{n_x-3}^{n+1} \\ T_{n_x-2}^{n+1} \end{bmatrix} = \begin{bmatrix} f_1^n \\ f_2^n \\ f_3^n \\ \dots \\ f_{n_x-4}^n \\ f_{n_x-3}^n \\ f_{n_x-2}^n \end{bmatrix}. \quad (2)$$

If we can achieve this, we can readily obtain the T^{n+1} values. More specifically, from the matrix equation, we know

$$T_{n_x-2}^{n+1} = f_{n_x-2}^n. \quad (3)$$

Then, we can recursively find $T_{n_x-3}^{n+1}$ all the way to T_1^{n+1} by solving the equation

$$\begin{aligned} T_j^{n+1} + \beta_j T_{j+1}^{n+1} &= f_j^n, \\ T_j^{n+1} &= f_j^n - \beta_j T_{j+1}^{n+1}. \end{aligned} \quad (4)$$

Every term on the RHS of the above equation is known.

We are now left to work out how to manipulate Eq. (1) to Eq. (2). For the first row, we can see that we can achieve the transformation by

$$\beta_1 = \frac{c_1}{b_1}, \quad f_1^n = \frac{d_1^n}{b_1}. \quad (5)$$

For the second row, we can subtract

$$a_2 T_1^{n+1} + b_2 T_2^{n+1} + c_2 T_3^{n+1} = d_2^n$$

with the first row of the transformed matrix multiplied by a_2 :

$$a_2 \times (T_1^{n+1} + \beta_1 T_2^{n+1}) = a_2 \times (f_1^n),$$

to get

$$\begin{aligned} (b_2 - a_2 \beta_1) T_2^{n+1} + c_2 T_3^{n+1} &= d_2^n - a_2 f_1^n, \\ T_2^{n+1} + \frac{c_2}{b_2 - a_2 \beta_1} T_3^{n+1} &= \frac{d_2^n - a_2 f_1^n}{b_2 - a_2 \beta_1}. \end{aligned}$$

We can identify

$$\beta_2 = \frac{c_2}{b_2 - a_2 \beta_1}, \quad f_2^n = \frac{d_2^n - a_2 f_1^n}{b_2 - a_2 \beta_1}. \quad (6)$$

Following the steps, in general, we have

$$\beta_j = \frac{c_j}{b_j - a_j \beta_{j-1}}, \quad f_j^n = \frac{d_j^n - a_j f_{j-1}^n}{b_j - a_j \beta_{j-1}}. \quad (7)$$

Crank-Nicolson in 2D

Building on our previous lecture, we now discuss how to extend the Crank-Nicolson scheme to higher spatial dimensions. We do it here for 2D, but the same concept applies to 3D. The 2D heat conduction equation is

$$\frac{\partial T}{\partial t} = \kappa \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right), \quad (8)$$

and we may write the Crank-Nicolson scheme as

$$\begin{aligned} \frac{T_{j,l}^{n+1} - T_{j,l}^n}{\Delta t} &= \frac{\kappa}{2} \left(\frac{T_{j+1,l}^n - 2T_{j,l}^n + T_{j-1,l}^n}{\Delta x^2} + \frac{T_{j+1,l}^{n+1} - 2T_{j,l}^{n+1} + T_{j-1,l}^{n+1}}{\Delta x^2} \right) \\ &+ \frac{\kappa}{2} \left(\frac{T_{j,l+1}^n - 2T_{j,l}^n + T_{j,l-1}^n}{\Delta y^2} + \frac{T_{j,l+1}^{n+1} - 2T_{j,l}^{n+1} + T_{j,l-1}^{n+1}}{\Delta y^2} \right). \end{aligned} \quad (9)$$

At this point, it is useful to introduce some shorthands. We will define $r_x = \frac{\kappa \Delta t}{\Delta x^2}$ and $r_y = \frac{\kappa \Delta t}{\Delta y^2}$. Following the same manipulation as in 1D, we get a system of linear equations given by

$$\begin{aligned} -r_x T_{j-1,l}^{n+1} - r_y T_{j,l-1}^{n+1} + 2(1 + r_x + r_y) T_{j,l}^{n+1} - r_x T_{j+1,l}^{n+1} - r_y T_{j,l+1}^{n+1} &= \\ r_x T_{j-1,l}^n + r_y T_{j,l-1}^n + 2(1 - r_x - r_y) T_{j,l}^n + r_x T_{j+1,l}^n + r_y T_{j,l+1}^n \end{aligned} \quad (10)$$

or, if we write using the k -index (recall our Lecture 4; with $k = ln_x + j$):

$$-r_x T_{k-1}^{n+1} - r_y T_{k-n_x}^{n+1} + 2(1 + r_x + r_y) T_k^{n+1} - r_x T_{k+1}^{n+1} - r_y T_{k+n_x}^{n+1} = d_k^n. \quad (11)$$

The right hand side of the equation consists of terms we already know at time step n :

$$d_k^n = r_x T_{k-1}^n + r_y T_{k-n_x}^n + 2(1 - r_x - r_y) T_k^n + r_x T_{k+1}^n + r_y T_{k+n_x}^n.$$

If we construct a matrix equation to represent Eq. (11) you can immediately see that this will not be a tri-diagonal matrix anymore. The matrix has a structure as follows:

$$\begin{bmatrix} \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & -r_y & \dots & -r_x & 2(1 + r_x + r_y) & -r_x & \dots & -r_y & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{bmatrix} \begin{bmatrix} \dots \\ T_{k-n_x}^{n+1} \\ \dots \\ T_{k-1}^{n+1} \\ T_k^{n+1} \\ T_{k+1}^{n+1} \\ \dots \\ T_{k+n_x}^{n+1} \\ \dots \end{bmatrix} = \begin{bmatrix} \dots \\ d_{k-n_x}^n \\ \dots \\ d_{k-1}^n \\ d_k^n \\ d_{k+1}^n \\ \dots \\ d_{k+n_x}^n \\ \dots \end{bmatrix}. \quad (12)$$

This means we cannot use the Thomas algorithm. We can employ a dense matrix method (e.g. from LAPACK), a direct sparse matrix method (e.g. LU factorisation), or an iterative sparse matrix solver (e.g. GMRES). Alternatively, we can further manipulate the matrix equation to make it a tri-diagonal system. This can be achieved using the Alternating Direction Implicit (ADI) method.

Alternating Direction Implicit (ADI) Method

The core idea behind the ADI scheme is to apply the Crank-Nicholson scheme in one direction at a time, such that we preserve the tri-diagonal structure. This approach leads to the use of a two step method, where each step evolves the solution over half a time step. More specifically:

$$\begin{aligned} \text{step 1 : } \frac{T_{j,l}^{n+\frac{1}{2}} - T_{j,l}^n}{\Delta t/2} &= \kappa \left(\frac{T_{j+1,l}^{n+\frac{1}{2}} - 2T_{j,l}^{n+\frac{1}{2}} + T_{j-1,l}^{n+\frac{1}{2}}}{\Delta x^2} + \frac{T_{j,l+1}^n - 2T_{j,l}^n + T_{j,l-1}^n}{\Delta y^2} \right) \\ \text{step 2 : } \frac{T_{j,l}^{n+1} - T_{j,l}^{n+\frac{1}{2}}}{\Delta t/2} &= \kappa \left(\frac{T_{j+1,l}^{n+\frac{1}{2}} - 2T_{j,l}^{n+\frac{1}{2}} + T_{j-1,l}^{n+\frac{1}{2}}}{\Delta x^2} + \frac{T_{j,l+1}^{n+\frac{1}{2}} - 2T_{j,l}^{n+\frac{1}{2}} + T_{j,l-1}^{n+\frac{1}{2}}}{\Delta y^2} \right) \end{aligned}$$

The terms in blue are known at the respective steps, while the terms in red are unknown and have to be solved simultaneously. Importantly, each step can be constructed as a tri-diagonal matrix equation and be solved with the Thomas algorithm. ADI is an example of an operator splitting method that is quite common for solving higher dimension PDEs.

Exercise 1: Write down steps 1 and 2 in terms of tri-diagonal matrix equations. State any assumptions you make.

Introduction to Hyperbolic PDEs

The last type of PDEs to be considered in this course is hyperbolic PDEs. The model second order linear hyperbolic PDE is the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad (13)$$

where u is the amplitude of the wave and c is the wave velocity. For simplicity, however, for the numerical part of the course we will focus on a simpler first order linear hyperbolic PDE corresponding to the advection (or first order wave) equation

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0. \quad (14)$$

We do this because (i) the advection equation is important for many transport problems, and (ii) we can capture the complexity of numerical approach for hyperbolic PDEs with this simpler equation. Formally we can write the second order wave equation in terms of coupled first order linear equation:

$$\begin{aligned}\frac{\partial u}{\partial t} - c \frac{\partial v}{\partial x} &= v, \\ \frac{\partial v}{\partial t} + c \frac{\partial u}{\partial x} &= 0.\end{aligned}$$

Hyperbolic PDEs arise in physical applications as models of waves, such as acoustic, elastic, electromagnetic, or gravitational waves.

Advection Equation

The prototypical equation we will use to discuss numerical methods for hyperbolic PDEs is the advection equation:

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0. \quad (15)$$

We can see immediately that this has the solution of the form $u(x - ct)$. This has a simple interpretation in that the function u is advected with a velocity c with time.

At this point it is useful to compare the typical evolution resulting from a diffusion and an advection equation, as shown in Fig. 1. In terms of difficulty in solving these PDEs numerically, a diffusion equation is simpler because the solution tends to become smoother with time. In contrast, for an advection equation, any sharp feature such as a step function is preserved. This often causes many difficulties in solving hyperbolic PDEs numerically. We will discuss this further in the next lecture.

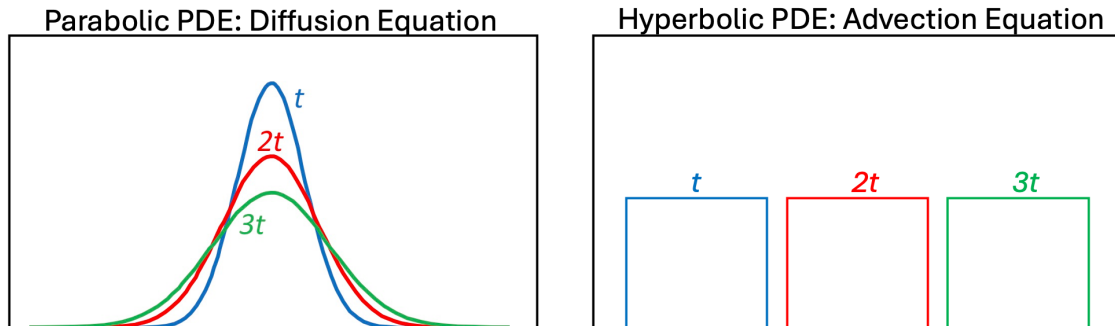


Figure 1: Typical evolution of a diffusion (left) and an advection (right) equations. In the diffusion equation, as time progresses, the feature of the function tends to become smoother. In contrast, in an advection equation, the feature complexity remains as the function is propagated in time.

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